CMAP: Continuous Masked Action-Tape Modeling for Visuomotor Manipulation

Anonymous Authors

***Abstract*— Action-chunking policies generate temporally coherent action sequences but usually treat a predicted chunk as a monolithic output, so a short contact or alignment error can compromise an otherwise valid motion. We propose CMAP, a history-conditioned continuous masked action policy that treats each chunk as an editable prediction. CMAP combines recent executed commands and future action positions in a continuous action tape. Given visual observations, robot proprioception, and executed-action history, a coarse decoder predicts a K-step future chunk and span-selection scores. A max-mean rule selects a high-scoring contiguous interval, and a local refiner regenerates the selected interval while preserving coarse actions outside it. The merged chunk is executed with receding-horizon control, repeatedly anchoring future prediction to commands already issued by the robot. CMAP averages 63.9% on eight RoboTwin tasks, compared with 45.8% for ACT and 34.8% for Diffusion Policy, and reaches 53.3% on ManiSkill versus 26.0% for ACT. Across AGIBOT G1 and Leju Kuavo 4 Pro evaluations, CMAP completes 38 of 50 outcomes, compared with 25 for ACT and 24 for Diffusion Policy. These results demonstrate consistent gains from history-conditioned local regeneration in simulated and physical visuomotor manipulation.**

***Index Terms*— Imitation Learning, Machine Learning for Robot Control, Action Chunking, Masked Generation**

I. Introduction

Visuomotor imitation learning requires a policy to translate high-dimensional observations and robot state into actions that remain coherent over time. Single-step control provides frequent feedback but can amplify prediction noise, whereas action chunking jointly predicts a short horizon and has proved effective across a broad range of manipulation tasks [1], [2]. The resulting chunk supplies a temporally coordinated motion plan, but it is commonly produced and consumed as one prediction. Errors near short contact or alignment transitions can then invalidate the critical part of a chunk even when the surrounding actions remain useful. As illustrated in Figure 1, conventional action-chunking methods generate the future sequence as a whole, whereas CMAP uses executed-action history to revise selected spans and preserve the rest of the prediction.

Generative policies represent richer action distributions through continuous diffusion, latent actions, or flow-style sequence models [2], [3], [4], [5], [6]. Masked generation makes these predictions editable: MGP selectively remasks discrete action tokens, related masked VLA methods refine tokens in parallel, and continuous action-inpainting preserves committed actions during asynchronous execution [7], [8], [9], [10], [11], [12]. Local revision in native continuous space must therefore solve two coupled problems: identify *where* revision is useful and reconstruct that interval consistently with both surrounding predictions and actions already executed.

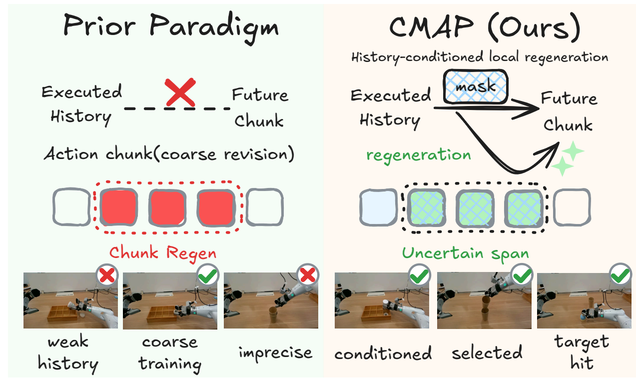

Fig. 1. CMAP versus conventional action-chunk prediction. CMAP uses executed history to select and regenerate uncertain future spans while preserving the rest of the chunk.

Executed commands provide temporal information that the current observation does not explicitly encode. Under receding-horizon control, latency, contact, and tracking error can make the realized transition differ from the previously predicted motion. A short executed-action history anchors the next chunk to the trajectory already sent to the robot, allowing a local refiner to preserve reliable future portions and reopen selected coherent intervals.

We introduce **CMAP**, a history-conditioned continuous masked action policy built around this principle. CMAP concatenates H executed actions and K future positions into a continuous action tape. Visual observations and proprioception form the observation context, while action embeddings represent visible history. A coarse decoder predicts the future chunk and temporal features for an action head and a span-selection score head. CMAP averages the score of every length-L candidate window, selects the maximizing interval $\mathcal{R}$, and regenerates that masked interval with a local refiner. This selection and regeneration process runs for P passes, preserving actions outside the selected intervals. The merged chunk is then executed with receding-horizon control. The pipeline separates span localization from continuous reconstruction while sharing observation and history context.

Training supervises coarse, regenerated, and merged actions plus span scores. RoboTwin uses $H = 4$, $K = 50$, and $(\rho, P, L) = (0, 1, 8)$, producing one 8-step regeneration interval. ManiSkill uses $K = 30$ and $(P, L) = (1, 8)$.

Experiments test CMAP in simulation and on physical robots. It averages 63.9% on eight RoboTwin tasks and 53.3% on ManiSkill, exceeding matched ACT and Diffusion

Policy baselines. Component ablations isolate the gains from executed history, masked regeneration, and score-guided selection. On AGIBOT G1 and Leju Kuavo 4 Pro, CMAP completes 38 of 50 outcomes, compared with 25 for ACT and 24 for Diffusion Policy.

To summarize, our main contributions are threefold:

- **Editable continuous action chunks.** A predict–select–regenerate framework that revises targeted intervals while preserving valid actions outside them.
- **History-conditioned action tape.** A continuous masked representation that anchors coarse prediction and local regeneration to recently executed commands.
- **Score-guided local regeneration.** A learned temporal score selects a contiguous future interval for local regeneration, with consistent gains across RoboTwin, ManiSkill, AGIBOT G1, and Kuavo 4 Pro.

II. Related Work

A. Visuomotor Action Chunking and Generative Policies

Imitation learning policies map visual observations and robot state to actions from demonstrations [13], [14]. Action chunking has become a strong design pattern because predicting multiple future actions jointly can improve temporal consistency and reduce the burden of single-step feedback [1], [15]. Generative policies extend this idea by modeling action chunks with richer conditional distributions: Diffusion Policy (DP) performs iterative denoising in continuous action space [2], VQ-BeT uses latent action abstractions [3], and recent large policies use diffusion or flow-style action generation for broader robot control [4], [5], [6]. These methods improve the global quality and expressiveness of action chunks. CMAP complements this direction through selective revision of local errors inside a predicted chunk.

B. Masked Generation and Action Inpainting

Masked prediction has been widely adopted for conditional sequence completion and iterative refinement, as exemplified by Mask-Predict and MaskGIT [16], [17]. This paradigm was subsequently extended to sequential decision making through masked trajectory modeling [18], [19]. Recent robot policies further employ masked or discrete-diffusion decoding for parallel action generation and iterative refinement [7], [8], [9], [10]. Continuous action-inpainting methods use partial action conditioning for asynchronous execution [11], [12]. CMAP uses partial conditioning to regenerate a future interval while preserving unselected actions outside it during execution.

C. Selective Revision and Score-Guided Refinement

Selective refinement requires identifying which parts of a prediction are most in need of revision. Prior work has explored uncertainty estimation through heteroscedastic prediction, ensembles, and stochastic inference to characterize model confidence [20], [21], [22]. In robot learning, related reliability signals support failure detection and intervention during policy execution [23], [24], [25]. CMAP learns a score to localize a regeneration interval in each predicted chunk.

III. Method

Figure 2 summarizes CMAP's predict–select–regenerate pipeline. CMAP constructs an action tape by combining H executed actions with K future action positions. It first predicts a coarse future chunk and a per-step span-selection score v_i. For every contiguous window of length L, CMAP computes the mean score s_j and selects the maximizing window as $\mathcal{R}$. The selected span is masked and regenerated, while all positions outside $\mathcal{R}$ retain their current predictions. CMAP repeats this operation for P passes and excludes previously regenerated future positions from later selections. The completed action tape is then executed with receding-horizon control during each closed-loop policy update.

A. Problem Setup

We consider visuomotor imitation learning from demonstrations. At timestep t, the policy observes visual inputs I_t, robot proprioception q_t, and a short history of executed actions. The goal is to output a future action chunk

$$\mathbf{a}_t^{\mathrm{fut}} = [a_t, a_{t+1}, \ldots, a_{t+K-1}] \in \mathbb{R}^{K \times d_a}, \tag{1}$$

where K is the chunk length and d_a is the action dimension. Let the recent action history be

$$\mathbf{a}_t^{\mathrm{hist}} = [a_{t-H}, \ldots, a_{t-1}] \in \mathbb{R}^{H \times d_a}. \tag{2}$$

The full CMAP instantiation concatenates the two parts into an action tape

$$\mathbf{x}_t = \left[\mathbf{a}_t^{\mathrm{hist}} \mid \mathbf{a}_t^{\mathrm{fut}}\right] \in \mathbb{R}^{(H+K) \times d_a}. \tag{3}$$

The tape is a conditional prediction object: visible history positions provide context, while masked future positions define the policy query used to predict future actions.

We denote a binary mask by $\mathbf{m} \in \{0,1\}^{H+K}$, where $m_i = 1$ indicates a hidden position. The main policy query keeps all valid history positions visible and masks all valid future positions. ManiSkill instead projects the last H actions linearly into the state embedding; both benchmarks retain the same future-chunk prediction and span-regeneration stages.

B. Continuous Masked Action-Tape Modeling

CMAP embeds each RoboTwin tape position with a position query and either a visible action embedding or a learned mask embedding. For a visible action x_i, the input token is

$$\mathbf{z}_i = \mathbf{e}_i + \phi_a(x_i), \tag{4}$$

where $\mathbf{e}_i$ is a learned positional query and ϕ_a projects continuous actions into the model dimension. For a masked position, the input token is

$$\mathbf{z}_i = \mathbf{e}_i + \mathbf{e}_{\mathrm{mask}}. \tag{5}$$

A visual encoder extracts observation features from I_t, proprioceptive features are computed from q_t, and a transformer predicts continuous future actions conditioned on these features and the available history representation. Following ACT [1], a latent encoder produces $q_\phi(z \mid \cdot) =$

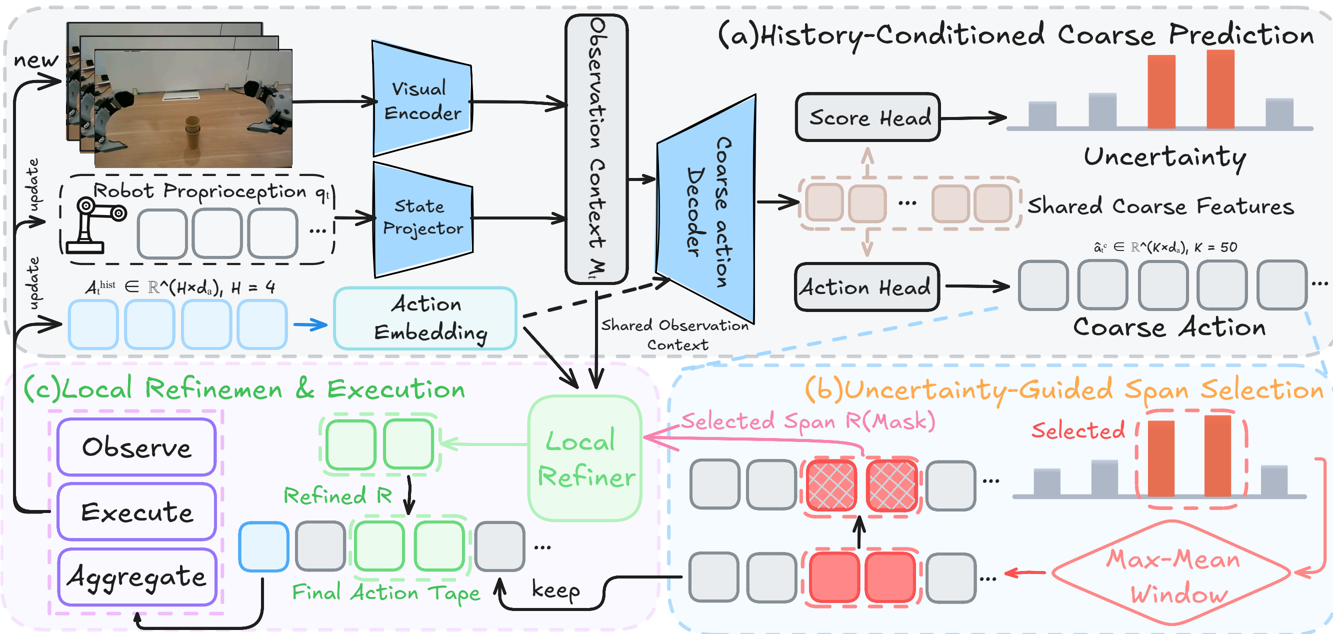

Fig. 2. CMAP pipeline. (a) Observation and executed history condition a coarse K-step prediction and span scores. (b) The highest-scoring contiguous window defines $\mathcal{R}$. (c) The local refiner regenerates $\mathcal{R}$ and preserves other actions; selection and regeneration repeat for P passes before receding-horizon execution.

$\mathcal{N}(\mu_z, \mathrm{diag}(\sigma_z^2))$. In the RoboTwin implementation, the latent encoder receives visible history and zero-filled future positions. The regularizer is

$$\mathcal{L}_{\mathrm{KL}} = -\frac{1}{2}\sum_j \left(1 + \log\sigma_{z,j}^2 - \mu_{z,j}^2 - \sigma_{z,j}^2\right), \tag{6}$$

At inference, CMAP sets the latent variable to zero.

C. Masked Training and Objective

Training deterministically masks all valid future positions and keeps history visible in the main configuration. Let $\hat{\mathbf{a}}^c$ be the coarse chunk, $\hat{\mathbf{a}}^r$ the regenerated prediction on selected span $\mathcal{R}$, and $\hat{\mathbf{a}}^f$ the final chunk after replacement. For N valid future steps, the losses are

$$\begin{aligned}
\mathcal{L}_{\mathrm{coarse}} &= \frac{1}{N}\sum_i \|\mathbf{a}_i - \hat{\mathbf{a}}_i^c\|_1, \\
\mathcal{L}_{\mathrm{regen}} &= \frac{1}{|\mathcal{R}|}\sum_{i\in\mathcal{R}} \|\mathbf{a}_i - \hat{\mathbf{a}}_i^r\|_1, \\
\mathcal{L}_{\mathrm{final}} &= \frac{1}{N}\sum_i \|\mathbf{a}_i - \hat{\mathbf{a}}_i^f\|_1.
\end{aligned} \tag{7}$$

The losses are normalized over valid time steps, and each ℓ_1 norm sums across all action dimensions. RoboTwin predicts a clipped log-variance $\ell_i \in [-5, 5]$ from a stop-gradient decoder state and supervises it with the stop-gradient coarse squared error,

$$e_i^{\mathrm{RT}} = \frac{1}{d_a}\|\mathbf{a}_i - \mathrm{sg}(\hat{\mathbf{a}}_i^c)\|_2^2, \quad \mathcal{L}_{\mathrm{unc}}^{\mathrm{RT}} = \frac{1}{N}\sum_i \left[e^{-\ell_i} e_i^{\mathrm{RT}} + \ell_i\right]. \tag{8}$$

We use this quantity to score candidate refinement spans. ManiSkill predicts $u_i = \mathrm{softplus}(Wh_i)$ and regresses the normalized coarse ℓ_1 error,

$$e_i^{\mathrm{MS}} = \frac{1}{d_a}\|\mathrm{sg}(\hat{\mathbf{a}}_i^c) - \mathbf{a}_i\|_1, \quad \mathcal{L}_{\mathrm{unc}}^{\mathrm{MS}} = \mathrm{MSE}(u_i, e_i^{\mathrm{MS}}). \tag{9}$$

The benchmark-specific objectives are

$$\begin{aligned}
\mathcal{L}_{\mathrm{RT}} &= \mathcal{L}_{\mathrm{coarse}} + \mathcal{L}_{\mathrm{regen}} + \mathcal{L}_{\mathrm{final}} + \mathcal{L}_{\mathrm{unc}}^{\mathrm{RT}} + 10\mathcal{L}_{\mathrm{KL}}, \\
\mathcal{L}_{\mathrm{MS}} &= \tfrac{1}{3}(\mathcal{L}_{\mathrm{coarse}} + \mathcal{L}_{\mathrm{regen}} + \mathcal{L}_{\mathrm{final}}) + 0.1\mathcal{L}_{\mathrm{unc}}^{\mathrm{MS}} + 10\mathcal{L}_{\mathrm{KL}}.
\end{aligned} \tag{10}$$

D. Selective Span Regeneration

After coarse prediction, CMAP assigns each future position a span-selection score v_i, using $v_i = \ell_i$ on RoboTwin and $v_i = u_i$ on ManiSkill. For span length L, it scores every contiguous candidate window by

$$s_j = \frac{1}{L}\sum_{i=j}^{j+L-1} v_i, \qquad j^\star = \arg\max_j s_j. \tag{11}$$

The first maximizing window is retained when scores tie. Let $\hat{a}^{(0)} = \hat{a}^c$. At pass p, CMAP selects $\mathcal{R}_p = \{j_p^\star, \ldots, j_p^\star + L - 1\}$, masks that span, and updates only the selected positions:

$$\hat{a}_i^{(p)} = \begin{cases} \hat{a}_i^{r,(p)}, & i \in \mathcal{R}_p, \\ \hat{a}_i^{(p-1)}, & i \notin \mathcal{R}_p. \end{cases} \tag{12}$$

Previously regenerated future positions are excluded from later selections. After P passes, the final chunk is $\hat{a}^f = \hat{a}^{(P)}$, and every position outside $\bigcup_{p=1}^{P}\mathcal{R}_p$ retains its coarse prediction.

IV. Simulation Experiments

A. Benchmarks and Evaluation Protocols

We evaluate CMAP on eight RoboTwin tasks [33] and three single-arm ManiSkill tasks [34], using 50 and 100

TABLE I

ROBOTWIN TASK SUCCESS RATES (%). AVG. IS THE EIGHT-TASK MACRO-AVERAGE. BEST VALUES AMONG ACT, DP, AND CMAP ARE BOLD, AND CMAP IS SHADED.

Method	Beat Block Hammer	Pick Dual Bottles	Handover Mic	Stack Two Blocks	Open Microwave	Click Alarm Clock	Open Laptop	Place Empty Cup	Avg.
RDT-1B [5]	77.0	42.0	90.0	21.0	37.0	61.0	59.0	56.0	55.4
TwinVLA [26]	77.0	18.0	84.0	26.0	3.0	33.0	80.0	50.0	46.4
ACT [1]	45.0	20.0	83.0	19.0	65.0	29.0	60.0	45.0	45.8
SeedPolicy-T [27]	72.0	24.0	92.0	47.0	80.0	61.0	45.0	32.0	56.6
Diffusion Policy [2]	42.0	24.0	53.0	7.0	5.0	61.0	49.0	37.0	34.8
CMAP (Ours)	**65.0**	**26.0**	**97.0**	**23.0**	**92.0**	**74.0**	**78.0**	**56.0**	**63.9**

TABLE II

MANISKILL SUCCESS RATES (%). BOLD COMPARES ACT AND CMAP; † DENOTES RESULTS REPORTED BY STAMO [28].

Method	Push Cube	Stack Cube	Pull Cube	Avg.
OpenVLA [29]†	4.0	18.0	16.0	12.7
UniVLA [30]†	7.0	22.0	20.0	16.3
WorldVLA [31]†	10.0	17.0	22.0	16.3
ACT [1]	31.0	13.0	34.0	26.0
StaMo [28]†	32.0	16.0	38.0	28.7
OpenVLA + StaMo state [29], [28]†	32.0	48.0	36.0	38.7
DINO-WM [32]†	8.0	4.0	18.0	10.0
CMAP (Ours)	**54.0**	**18.0**	**88.0**	**53.3**

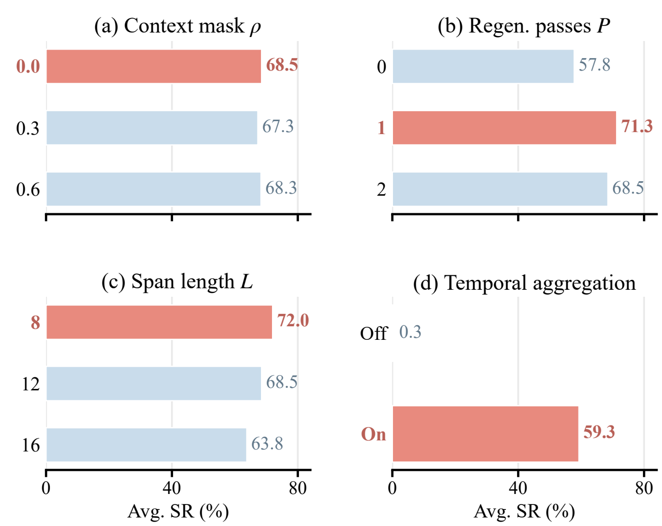

Fig. 3. CMAP sensitivity on four RoboTwin tasks. Coral bars mark selected settings; each bar is the four-task macro-average.

expert demonstrations per task, respectively. Figure 6 shows the RoboTwin tasks: *Beat Block Hammer*, *Pick Dual Bottles*, *Handover Mic*, *Stack Two Blocks*, *Open Microwave*, *Click Alarm Clock*, *Open Laptop*, and *Place Empty Cup*. They span tool contact, coordinated pickup, handover, alignment, articulated objects, and placement. The bottom row shows CMAP rollouts for ManiSkill *Push Cube*, *Stack Cube*, and *Pull Cube*. RoboTwin baselines are ACT [1], Diffusion Policy (DP) [2], RDT-1B [5], TwinVLA [26], and SeedPolicy-T [27]. ACT, DP, and CMAP are evaluated over 100 episodes per task; the remaining rows use published estimates.

B. Performance Comparison

a) RoboTwin Main Comparison: Table I summarizes the RoboTwin results. Among ACT, DP, and CMAP, CMAP achieves the highest success rate on all eight tasks and raises the macro-average to 63.9%, compared with 45.8% for ACT and 34.8% for DP. It also exceeds the strongest published macro-average in the table, 56.6% from SeedPolicy-T. CMAP reaches 97.0% on *Handover Mic*, 74.0% on *Click Alarm Clock*, and 78.0% on *Open Laptop*, demonstrating strong performance across coordinated transfer, contact triggering, and articulated-object manipulation.

b) ManiSkill Comparison: Table II summarizes the ManiSkill results. Boldface compares ACT and CMAP; CMAP is higher on all three tasks and the overall average. CMAP records 54.0%, 18.0%, and 88.0% on *Push Cube*, *Stack Cube*, and *Pull Cube*, respectively, averaging 53.3%.

C. Ablation Studies

a) Core Components: Figure 4 compares four configurations across RoboTwin and ManiSkill. ACT is the base policy; SeqHist adds executed-action history; RandSpan adds contiguous masked regeneration with random span placement; and CMAP replaces random placement with score-guided span selection. RandSpan and CMAP use the one-pass, 8-step RoboTwin regeneration budget, while CMAP uses $\rho = 0$ and score-guided placement.

Figure 5 expands the comparison to six configurations on three representative tasks and their overall average. Refine-Block performs residual refinement over a contiguous block, while Refine-Uncert refines individually selected high-uncertainty steps. RandSpan and CMAP instead regenerate masked spans using random and score-guided selection, respectively. CMAP annotations show gains over the best alternative in every reported group.

b) Configuration Sensitivity: Figure 3 varies context masking, regeneration passes, span length, and temporal aggregation on *Beat Block Hammer*, *Open Microwave*, *Open Laptop*, and *Click Alarm Clock*. Each training sweep changes one parameter from the reference $\rho = 0$, $P = 1$, and $L = 12$, and reports the four-task macro-average from validation-selected checkpoints. The sweeps select $\rho = 0$, $P = 1$, and $L = 8$, with average success rates of 68.5%, 71.3%, and

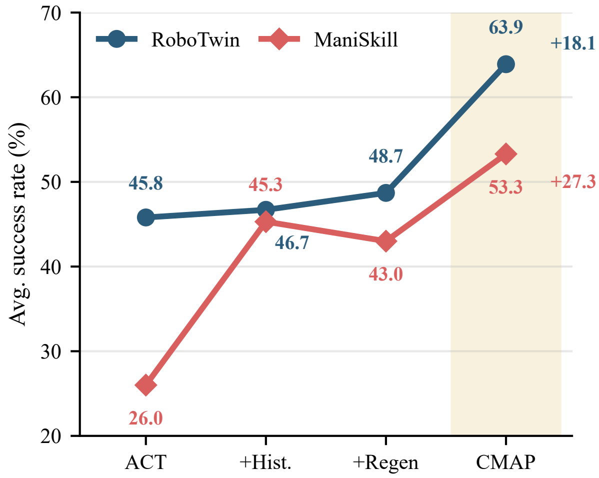

Fig. 4. Component ablations on RoboTwin and ManiSkill. +Hist. adds executed history, +Regen masked span regeneration, and CMAP score-guided selection; CMAP labels show gains over ACT.

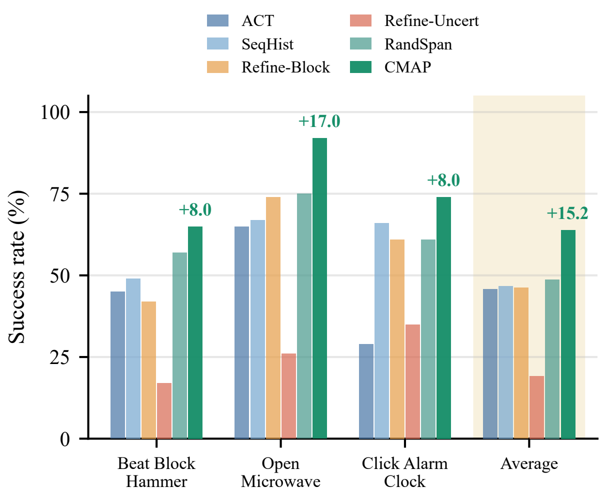

Fig. 5. RoboTwin task-level ablations and overall average; labels show gains over the best alternative.

72.0%, respectively. The execution sweep fixes the 6000-iteration CMAP checkpoint and per-step replanning ($h = 1$), then toggles temporal aggregation. Aggregating overlapping action chunks raises the average success rate from 0.3% to 59.3%. RoboTwin therefore uses $(\rho, P, L) = (0, 1, 8)$ with temporal aggregation enabled at $h = 1$.

c) Cross-Benchmark Ablation Summary: Figure 4 compares the four internal configurations on RoboTwin and ManiSkill. On RoboTwin, ACT, SeqHist, RandSpan, and CMAP obtain average success rates of 45.8%, 46.7%, 48.7%, and 63.9%, respectively. The corresponding ManiSkill values are 26.0%, 45.3%, 43.0%, and 53.3%. RoboTwin regenerates one 8-action span within a 50-step chunk; ManiSkill regenerates one 8-action span within a 30-step chunk.

V. Real-World Experiments

A. Real Robot Benchmark

We evaluate ACT, DP, and CMAP on AGIBOT G1 and Leju Kuavo 4 Pro. AGIBOT controls a right 7-DoF arm and scalar gripper from head- and wrist-camera RGB streams,

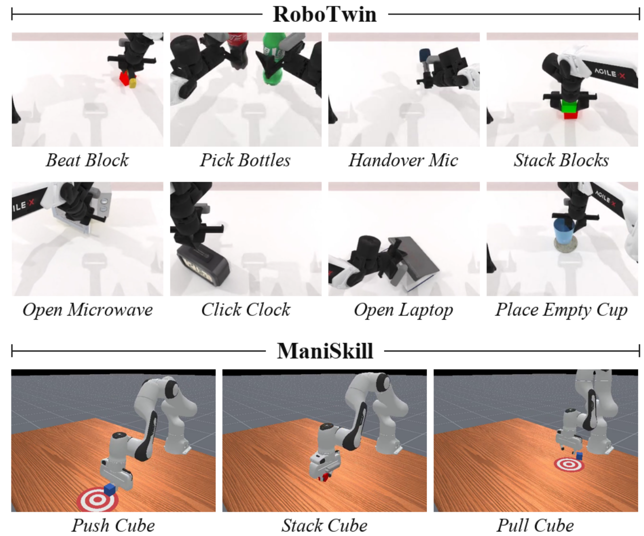

Fig. 6. Simulation tasks: RoboTwin (top) and ManiSkill (bottom).

using a 15-Hz policy loop and 100-Hz arm controller. Kuavo controls a 7-DoF arm with a dexterous hand from synchronized multi-view RGB at 10 Hz. These platforms test contact, arm–hand coordination, sensing noise, latency, and communication delay. Figures 7 and 8 show representative rollouts, and Table III reports the results.

a) Kuavo 4 Pro Tasks: Kuavo evaluates long-horizon *Sheet Loading*. The robot must grasp and transport a sheet-metal workpiece, align it with the target fixture, and release it while maintaining coordinated arm–hand control. Figure 7 shows initialization, pre-grasp alignment, grasp and lift, and placement.

b) AGIBOT G1 Tasks: AGIBOT evaluates three tabletop tasks: *Cup Separation*, *Bell Press*, and *Dual-Item Placement*. The last task places a chess piece and bottle into separate box compartments and reports two sequential checkpoints, *Chess Grasp* and *Bottle Grasp*. The suite covers short contact, precise triggering, and multi-object progression. In Figure 8, columns are tasks and rows are successive execution stages.

B. Deployment Protocol

Each task uses 50 teleoperated demonstrations shared across ACT, DP, and CMAP. Logs contain synchronized RGB, proprioception, action commands, and timestamps. Each method is evaluated over 10 trials per task with matched observations and action spaces, and outcomes are verified from video and logs. *Chess Grasp* and *Bottle Grasp* are paired checkpoints in the same 10 *Dual-Item Placement* trials.

Both platforms enforce joint/workspace limits, joint-delta clipping, emergency stops, and home-reset procedures throughout every recorded physical trial on both platforms.

C. Performance Comparison

Table III reports success counts for all three methods. On AGIBOT G1, the aggregate rates over the four reported task or checkpoint outcomes are 57.5% for ACT, 57.5% for DP, and 85.0% for CMAP.

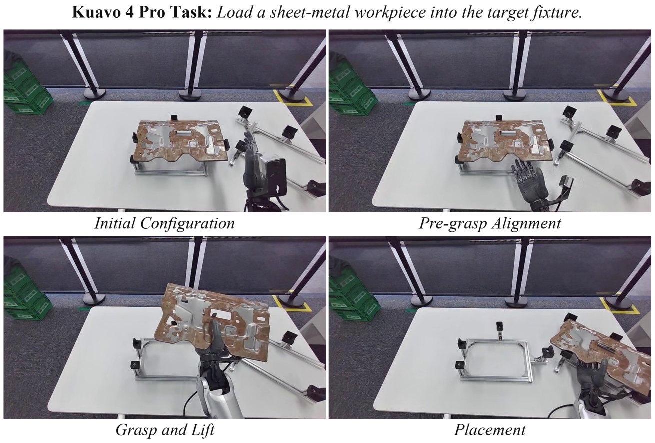

Fig. 7. Kuavo 4 Pro *Sheet Loading* rollout: initialization, alignment, grasp and lift, and placement.

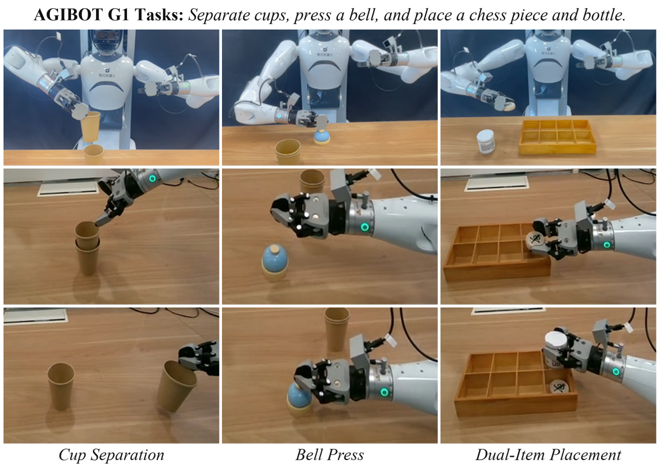

Fig. 8. AGIBOT G1 rollouts. Columns are tasks; rows are execution stages.

On Kuavo 4 Pro *Sheet Loading*, ACT, DP, and CMAP obtain 2/10, 1/10, and 4/10, respectively. Aggregating all five outcomes gives 25/50 (50.0%) for ACT, 24/50 (48.0%) for DP, and 38/50 (76.0%) for CMAP. The 50 matched outcomes combine four AGIBOT task or checkpoint measures with Kuavo *Sheet Loading*.

a) Cross-Platform Trends: CMAP improves aggregate AGIBOT success by 27.5 points over both ACT and DP, reaching 70.0% on *Cup Separation*, 90.0% on *Bell Press*, 80.0% on *Chess Grasp*, and 100.0% on *Bottle Grasp*. On Kuavo, it completes 4/10 *Sheet Loading* trials, compared with 2/10 for ACT and 1/10 for DP. The gains therefore span two robot embodiments and all reported task outcomes.

b) Execution-Level Interpretation: The tasks expose complementary correction demands: short contact in *Cup Separation* and *Bell Press*, ordered interactions in *Dual-Item Placement*, and sustained grasp during *Sheet Loading*. Executed-action history anchors each prediction to issued commands, while contiguous regeneration intervals preserve unaffected phases. The same mechanism therefore supports triggering, sequential grasping, and long-horizon transport.

c) Representative Execution Cases: Figure 9 pairs successful outcomes with four contact failures and one recovered near-failure on AGIBOT G1 and Kuavo 4 Pro. The columns show *Cup Separation*, *Bell Press*, *Chess Grasp*, *Bottle Grasp*, and *Sheet Loading*. Red callouts mark the critical contact regions.

TABLE III

SUCCESS COUNTS OVER 10 TRIALS; TOTAL SUMS FIVE OUTCOMES.

Method	**AGIBOT G1**				**Kuavo 4 Pro**	**Total**
	Cup Sep.	Bell Press	**Dual-Item Placement**		Sheet Load.	
			Chess Grasp	Bottle Grasp		
ACT	4/10	8/10	5/10	6/10	2/10	25/50
DP	5/10	7/10	3/10	8/10	1/10	24/50
CMAP	**7/10**	**9/10**	**8/10**	**10/10**	**4/10**	**38/50**

Cup Separation misses rim contact, while *Bell Press* applies force off center. During *Dual-Item Placement*, incomplete closure on the chess piece and a tilted bottle grasp undermine placement. *Sheet Loading* briefly relies on single-point support before recovering. These cases identify contact geometry, grasp closure, and object alignment as the main real-world failure sources.

The current policy scores future actions from visual observations, proprioception, and executed commands, without direct tactile or force measurements. The remaining failures identify a concrete extension path: contact sensing can augment the span-selection score, while a recovery controller can reopen a new interval when the expected contact event is absent. This combination would preserve CMAP's localized editing structure while improving detection and correction of contact-specific errors during closed-loop execution.

D. Discussion

a) Ablation-to-Deployment Connection: The simulation ablations clarify how CMAP's components contribute to the final policy. ACT, SeqHist, RandSpan, and CMAP obtain average success rates of 45.8%, 46.7%, 48.7%, and 63.9% on RoboTwin, respectively, and 26.0%, 45.3%, 43.0%, and 53.3% on ManiSkill. The task-level comparison further shows that CMAP exceeds the strongest alternative in all four reported groups. Together, these results connect the two central design decisions: executed history anchors the new prediction to realized control, and span scoring directs refinement toward the portion of the future tape most likely to benefit from revision.

b) Temporal Editing Granularity: CMAP deliberately edits a minority of each predicted chunk. The RoboTwin configuration regenerates 8 of 50 future actions in one pass, while the ManiSkill configuration regenerates 8 of 30 future actions. Positions outside the selected span preserve the coarse prediction and retain the global structure of the planned motion. In Figure 3, one regeneration pass achieves 71.3%, compared with 68.5% for two passes, while shortening the regenerated span from $L = 12$ to $L = 8$ raises the four-task average to 72.0%. At fixed $h = 1$, temporal aggregation raises the average from 0.3% to 59.3% by merging overlapping chunks when the policy replans at every step. These sweeps identify regeneration depth,

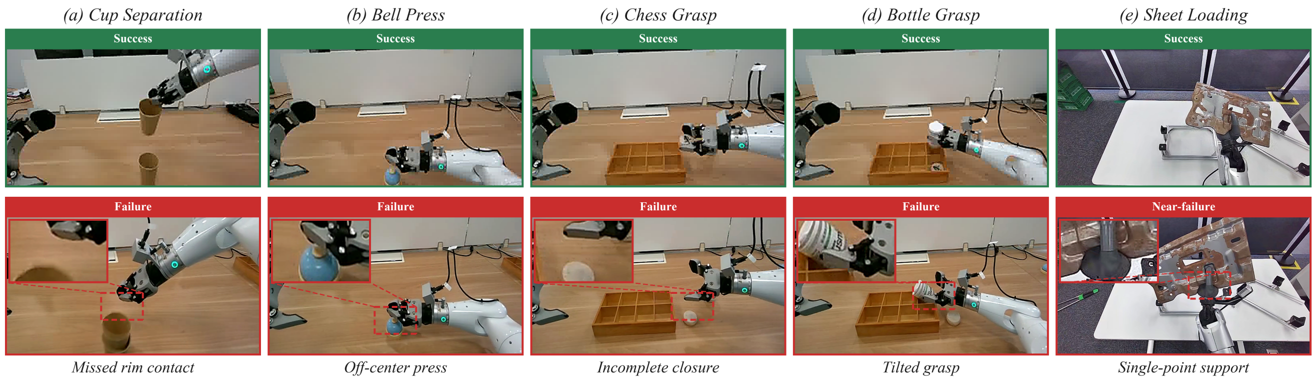

Fig. 9. Real-world execution pairs on AGIBOT G1 and Kuavo 4 Pro. Top: success. Bottom: four failures and one recovered near-failure. Red callouts mark critical contact regions.

span size, and action-chunk aggregation as complementary controls over localized prediction and execution.

c) Task-Dependent Behavior: The task-level results also identify where local regeneration is most effective. In the RoboTwin main comparison, CMAP leads the matched ACT and DP results on all eight tasks. Its gains over the stronger matched baseline are 27 points on *Open Microwave*, 20 on *Beat Block Hammer*, 18 on *Open Laptop*, 14 on *Handover Mic*, 13 on *Click Alarm Clock*, and 11 on *Place Empty Cup*. The smaller gains on *Pick Dual Bottles* and *Stack Two Blocks*, at 2 and 4 points, show that precision-heavy coordination remains demanding. On ManiSkill, the gains over ACT are 23 points on *Push Cube*, 5 points on *Stack Cube*, and 54 points on *Pull Cube*. The variation across tasks indicates that score-guided regeneration is especially valuable when a localized correction can recover the remainder of a motion.

d) Evaluation Scope and Extensions: The physical study uses matched training data and action spaces, 50 teleoperated demonstrations per task, and 10 trials for each reported task or checkpoint. This protocol isolates the policy comparison on the two robot platforms and exposes the methods to distinct camera layouts, control frequencies, end effectors, and contact patterns. A broader evaluation can extend this evidence through additional object instances, scene layouts, lighting conditions, and repeated training seeds. The diagnostic pairs further motivate contact-sensitive observations, adaptive span lengths, and event-triggered recovery passes. These extensions fit the existing action-tape formulation: new sensing channels can inform the score head, while the selected interval remains the unit of localized regeneration and closed-loop correction.

VI. Conclusion

We presented CMAP, a history-conditioned continuous masked action-tape framework for local revision of predicted action chunks. CMAP conditions coarse future prediction and local regeneration on recently executed actions, scores contiguous candidate spans, and regenerates the selected interval while preserving the remaining coarse prediction. CMAP obtains average success rates of 63.9% in the eight-task RoboTwin main comparison and 53.3% on ManiSkill. Controlled RoboTwin and ManiSkill ablations distinguish the contributions of visible action history, contiguous masked regeneration, and score-guided span selection. Across the five reported physical outcomes, CMAP achieves 38 successes in 50 trials, compared with 25 for ACT and 24 for DP. The task-level results show that localized regeneration benefits both short contact events and longer transport sequences, while the paired examples isolate contact establishment as the main remaining challenge. Future work will calibrate span-selection scores, incorporate contact-sensitive observations, adapt regeneration budgets, and broaden physical evaluation across additional robot platforms, task families, and contact conditions.

References

[1] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," in *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, July 2023.

[2] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. C. M. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, vol. 44, no. 10–11, pp. 1684–1704, 2025.

[3] S. Lee, Y. Wang, H. Etukuru, H. J. Kim, N. M. M. Shafiullah, and L. Pinto, "Behavior generation with latent actions," in *Proceedings of the 41st International Conference on Machine Learning*. PMLR, 2024.

[4] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, L. X. Shi, L. Smith, J. Tanner, Q. Vuong, A. Walling, H. Wang, and U. Zhilinsky, "π_0: A vision-language-action flow model for general robot control," in *Proceedings of Robotics: Science and Systems*, 2025.

[5] S. Liu, L. Wu, B. Li, H. Tan, H. Chen, Z. Wang, K. Xu, H. Su, and J. Zhu, "RDT-1B: A diffusion foundation model for bimanual manipulation," in *International Conference on Learning Representations*, 2025.

[6] G. Yan, J. Zhu, Y. Deng, S. Yang, R.-Z. Qiu, X. Cheng, M. Memmel, R. Krishna, A. Goyal, X. Wang, and D. Fox, "ManiFlow: A general robot manipulation policy via consistency flow training," in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 2268–2293. [Online]. Available: https://proceedings.mlr.press/v305/yan25a.html

[7] L. Zhuang, S. Fan, F. P. Audonnet, Y. Ru, E. S. Ho, G. Aragon-Camarasa, and P. Henderson, "Masked generative policy for robotic control," in *International Conference on Learning Representations*, 2026. [Online]. Available: https://arxiv.org/abs/2512.09101

[8] Y. Wen, H. Li, K. Gu, Y. Zhao, T. Wang, and X. Sun, "Llada-vla: Vision language diffusion action models," *arXiv preprint arXiv:2509.06932*, 2025.

[9] Z. Liang, Y. Li, T. Yang, C. Wu, S. Mao, L. Pei, T. Nian, S. Zhou, X. Yang, J. Pang *et al.*, "Discrete diffusion vla: Bringing discrete diffusion to action decoding in vision-language-action policies," *arXiv preprint arXiv:2508.20072*, 2025.

[10] S.-W. Lee, X. Kang, and Y.-L. Kuo, "Tbd-vla: Temporal block diffusion vision language action model," *arXiv preprint arXiv:2606.07895*, 2026.

[11] K. Black, M. Galliker, and S. Levine, "Real-time execution of action chunking flow policies," in *Advances in Neural Information Processing Systems*, vol. 38, 2025, pp. 33 383–33 407. [Online]. Available: https://arxiv.org/abs/2506.07339

[12] H. Wang, G. Zhang, Y. Yan, Y. Shang, R. Kompella, and G. Liu, "Real-time robot execution with masked action chunking," in *International Conference on Learning Representations*, 2026. [Online]. Available: https://arxiv.org/abs/2601.20130

[13] A. Mandlekar, D. Xu, J. Wong, S. Nasiriany, C. Wang, R. Kulkarni, L. Fei-Fei, S. Savarese, Y. Zhu, and R. Martín-Martín, "What matters in learning from offline human demonstrations for robot manipulation," in *Proceedings of the 5th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 164. PMLR, 2022, pp. 1678–1690.

[14] P. Florence, C. Lynch, A. Zeng, O. A. Ramirez, A. Wahid, L. Downs, A. Wong, J. Lee, I. Mordatch, and J. Tompson, "Implicit behavioral cloning," in *Proceedings of the 5th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 164. PMLR, 2022, pp. 158–168.

[15] Z. Fu, T. Z. Zhao, and C. Finn, "Mobile aloha: Learning bimanual mobile manipulation using low-cost whole-body teleoperation," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 4066–4083. [Online]. Available: https://proceedings.mlr.press/v270/fu25b.html

[16] M. Ghazvininejad, O. Levy, Y. Liu, and L. Zettlemoyer, "Mask-predict: Parallel decoding of conditional masked language models," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, 2019. [Online]. Available: https://aclanthology.org/D19-1633/

[17] H. Chang, H. Zhang, L. Jiang, C. Liu, and W. T. Freeman, "MaskGIT: Masked generative image transformer," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 11 305–11 315.

[18] M. Carroll, O. Paradise, J. Lin, R. Georgescu, M. Sun, D. Bignell, S. Milani, K. Hofmann, M. J. Hausknecht, A. D. Dragan, and S. Devlin, "UniMASK: Unified inference in sequential decision problems," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[19] P. Wu, A. Majumdar, K. Stone, Y. Lin, I. Mordatch, P. Abbeel, and A. Rajeswaran, "Masked trajectory models for prediction, representation, and control," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 37 607–37 623. [Online]. Available: https://proceedings.mlr.press/v202/wu23o.html

[20] A. Kendall and Y. Gal, "What uncertainties do we need in bayesian deep learning for computer vision?" in *Advances in Neural Information Processing Systems 30*, 2017. [Online]. Available: https://papers.nips.cc/paper_files/paper/2017/hash/2650d6089a6d640c5e85b2b88265dc2b-Abstract.html

[21] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems 30*, 2017.

[22] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," in *Proceedings of the 33rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 48. New York, New York, USA: PMLR, 2016, pp. 1050–1059. [Online]. Available: https://proceedings.mlr.press/v48/gal16.html

[23] K. Menda, K. R. Driggs-Campbell, and M. J. Kochenderfer, "EnsembleDAgger: A bayesian approach to safe imitation learning," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2019.

[24] R. Hoque, A. Balakrishna, E. Novoseller, A. Wilcox, D. S. Brown, and K. Goldberg, "ThriftyDAgger: Budget-aware novelty and risk gating for interactive imitation learning," in *Proceedings of the 5th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 164. PMLR, 2022, pp. 598–608.

[25] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez, P. Miller, R. Lee, P. Shah, R. A. Ambrus, H. Nishimura, and M. Itkina, "Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies," in *Proceedings of Robotics: Science and Systems*, 2025.

[26] H. Im, E. Jeong, A. Kolobov, J. Fu, and Y. Lee, "TwinVLA: Data-efficient bimanual manipulation with twin single-arm vision-language-action models," in *The Fourteenth International Conference on Learning Representations*, 2026. [Online]. Available: https://openreview.net/forum?id=jG9W6nAwVz

[27] Y. Gui, Y. Zhou, S. Cheng, X. Yuan, H. Fan, P. Cheng, and S. Liu, "SeedPolicy: Horizon scaling via self-evolving diffusion policy for robot manipulation," *arXiv preprint arXiv:2603.05117*, 2026. [Online]. Available: https://arxiv.org/abs/2603.05117

[28] M. Liu, J. Shu, H. Chen, Z. Li, C. Zhao, J. Yang, S. Gao, H. Chen, and C. Shen, "StaMo: Unsupervised learning of generalizable robot motion from compact state representation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 35 014–35 024.

[29] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, "OpenVLA: An open-source vision-language-action model," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 2679–2713. [Online]. Available: https://proceedings.mlr.press/v270/kim25c.html

[30] Q. Bu, Y. Yang, J. Cai, S. Gao, G. Ren, M. Yao, P. Luo, and H. Li, "UniVLA: Learning to act anywhere with task-centric latent actions," in *Proceedings of Robotics: Science and Systems*, Los Angeles, California, USA, June 2025.

[31] J. Cen, C. Yu, H. Yuan, Y. Jiang, S. Huang, J. Guo, X. Li, Y. Song, H. Luo, F. Wang, D. Zhao, and H. Chen, "WorldVLA: Towards autoregressive action world model," *arXiv preprint arXiv:2506.21539*, 2025.

[32] G. Zhou, H. Pan, Y. LeCun, and L. Pinto, "DINO-WM: World models on pre-trained visual features enable zero-shot planning," *arXiv preprint arXiv:2411.04983*, 2024.

[33] Y. Mu, T. Chen, Z. Chen, S. Peng, Z. Lan, Z. Gao, Z. Liang, Q. Yu, Y. Zou, M. Xu, L. Lin, Z. Xie, M. Ding, and P. Luo, "Robotwin: Dual-arm robot benchmark with generative digital twins," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 27 649–27 660.

[34] J. Gu, F. Xiang, X. Li, Z. Ling, X. Liu, T. Mu, Y. Tang, S. Tao, X. Wei, Y. Yao, X. Yuan, P. Xie, Z. Huang, R. Chen, and H. Su, "ManiSkill2: A unified benchmark for generalizable manipulation skills," in *International Conference on Learning Representations*, 2023. [Online]. Available: https://openreview.net/forum?id=b_CQDy9vrD1